

Review retrospective:
*cumulants as derivatives of $\log Z$, and the arc's first landed
dead-open*

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Abstract

A soundness review of `Laplace/OneD/AnharmonicCumulantsLogPartition.lean`: the cumulant-generating function route, identifying the connected cumulants $\kappa_2, \kappa_3, \kappa_4$ with the second, third and fourth derivatives of $\log Z$ at the origin, through the ladder identity $\text{iteratedDeriv}(n+1) K 0 = \text{iteratedDeriv}(n) M 0$ (since $K' = M$, the mean, near 0). Result: **a clean fidelity pass with one landed simplification** — a genuinely dead open `MeasureTheory`, removed under rebuild-arbitration. The fourth-cumulant coefficients $(-4, -3, +12, -6)$ and the ladder index shift were the focus of the fidelity check and are correct.

1 Setting

The anharmonic arc proved the connected cumulants κ_n of $u = -(tx)$ first as derivatives of the perturbed mean $M(h) = G_1(h)/G_0(h)$ (the FDT/susceptibility route). This file closes the narrative loop by tying them to the genuine cumulant generating function $K(h) = \log Z(h)$, $Z = G_0$: it is the superstructure whose foundation, `logPartition_hasDerivAt` (the first rung $K'(0) = Z'(0)/Z(0)$), was reviewed immediately before this in the log-partition-deriv review. Reviewing the two together checks the full CGF route top-to-bottom.

2 What was reviewed

Six public theorems: the general- h first derivative $K'(h_0) = G_1(h_0)/G_0(h_0)$ ($|h_0| < 1$); the eventual equality $\text{deriv } K =_{\mathcal{N}_0} M$; the ladder $\text{iteratedDeriv}(n+1) K 0 = \text{iteratedDeriv}(n) M 0$; and the three cumulant identities $\kappa_2 = \partial^2 \log Z$, $\kappa_3 = \partial^3 \log Z$, $\kappa_4 = \partial^4 \log Z$, each delegating its closed form to the already-reviewed per-order theorems (κ_2 , κ_3 , and the fourth-cumulant file).

3 Fidelity / soundness findings

Sound; GPT found no mismatch on any of the six results. The two places a CGF formalisation can go wrong were the focus:

- **The ladder index shift.** $\text{iteratedDeriv}(n+1) K 0 = \text{iteratedDeriv}(n) M 0$ is correct: `iteratedDeriv_succ'` peels the *outer* derivative to leave $\text{iteratedDeriv } n (\text{deriv } K) 0$, and `Filter.EventuallyEq.iteratedDeriv_eq` swaps $\text{deriv } K$ for M using the neighbourhood equality. No off-by-one — $\kappa_2, \kappa_3, \kappa_4$ land at derivative orders 2, 3, 4 exactly.

- **The fourth-cumulant coefficients.** The numerator $G_4G_0^3 - 4G_3G_1G_0^2 - 3G_2^2G_0^2 + 12G_2G_1^2G_0 - 6G_1^4$, over G_0^4 , is exactly the standard fourth-cumulant-in-raw-moments expansion $m_4 - 4m_3m_1 - 3m_2^2 + 12m_2m_1^2 - 6m_1^4$ cleared of denominators ($m_k = G_k/G_0$). Coefficients $(-4, -3, +12, -6)$ verified against the standard formula and against the delegated `anharmonic_fourth_cumulant`, whose numerator is byte-identical.

The docstring is accurate: the file genuinely proves $\kappa_2, \kappa_3, \kappa_4 = \partial^n \log Z$ — which is precisely the forward-reference the first-rung file pointed at, now fulfilled — and “the standard definition of the connected cumulants” is the right characterisation.

4 Simplifications applied

One landed: a dead open MeasureTheory. The file declared `open MeasureTheory Topology`, but nothing in its statements or proofs uses any unqualified `MeasureTheory` name — there is no bare $\int$ notation and no unqualified `Measure/volume`; all integration is hidden behind the already-defined `weightedPartition`. `Topology`, by contrast, is load-bearing (the $\mathcal{N}0$ neighbourhood filter in `deriv_logPartition_eventuallyEq`). Removed in a worktree, **rebuild-arbitrated** (baseline green 8273 jobs $\rightarrow$ edit $\rightarrow$ rebuild green 8273, no new warnings, all six signatures byte-identical), GPT-affirmed, and fast-forward-merged to `main`.

GPT’s other six findings were declined: inlining the used `hZ/hne` locals, a `Metric.ball_mem_nhds` shortcut, `simp` compressions of the ladder and the three κ -wrappers, and proof-local Z/G abbreviations. All are tactic-golf or readability-only — they remove no dead code, subsume no hand-rolled argument with a Mathlib lemma, and drop no hypothesis. Under the review programme’s standing discipline these lateral, build-break-prone compressions are not landed.

5 Disagreements and open questions

None on the mathematics.

6 Lessons

- **A dead open is a real simplification; tactic-golf is not.** This review and the preceding one drew the line cleanly: seven of GPT’s eight total simplification proposals across the two files were proof-golf and declined, while the single genuine dead-open was landed. The discriminator is whether the change removes something the file does not use (dead code/import/open — a real, lint-visible improvement) versus merely rewriting a working proof more tersely (no statement benefit, real build-break risk).
- **The build is the only reliable dead-open arbiter — but it is reliable.** Two earlier reviews in this arc saw the dead-open heuristic misfire in both directions; here Claude’s grep and GPT agreed the `MeasureTheory` open was dead, and the rebuild settled it definitively. When both the static check and the independent reviewer concur *and* the rebuild stays green with byte-identical signatures, a dead-open removal is safe to land.

7 Follow-ups

- `AnharmonicCumulantMomentForms` is the last unreviewed file of the one-dimensional anharmonic arc; reviewing it completes the sweep.